

EBMF Diagnostic: Does Survival Signal Exist in Merged TCGA + CPTAC?

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1 Motivation

Training SSBMF on the merged TCGA_PAAD (RNA-seq) + CPTAC (proteomics) cohort consistently produces all factor coefficients $\hat{\beta}_k = 0$: no factor is selected as prognostic under the Bayesian survival objective. Before debugging the model, a more basic question must be answered:

Is the absence of signal a data problem or a model problem?

Two explanations are plausible:

1. *Data problem.* The RNA-seq vs. proteomics platform difference dominates all variance in the merged matrix; batch structure consumes the gene-set, leaving no detectable survival signal regardless of the analysis method.
2. *Model problem.* Survival signal exists in the data, but SSBMF's objective weighting, prior aggressiveness, or CAVI convergence behavior prevents it from reaching the β coefficients.

This diagnostic distinguishes between the two by running **purely unsupervised factor analysis** — Empirical Bayes Matrix Factorization (EBMF via `flashier`) and PCA — on the same merged training matrix used for SSBMF, then asking whether any resulting factor is univariately associated with survival via a Cox proportional-hazards model.

2 Methods

Data. v2-preprocessed merged training matrix: TCGA_PAAD ($n = 144$) + CPTAC ($n = 129$), $p = 2,000$ genes. Preprocessing: intersect raw gene universes (17,900 genes), $\log_2(x+1)$ transform (RNA-seq only), quantile normalization across all samples jointly, top-2,000 by merged-matrix variance, per-subject rank transform.

EBMF. `flashier::flash()` with per-gene residual variance (`var_type = 2`) and a greedy add + backfit strategy (`greedy_Kmax = 20`). Loadings extracted via LDF decomposition with unit- L_2 -norm columns so all factors are on a comparable scale.

PCA. `prcomp(Y, rank. = 20, center = TRUE)`.

Survival association. For each factor k , a univariate Cox model $h(t) = h_0(t) \exp(\ell_k \gamma_k)$ is fit where ℓ_k is the n -vector of subject loadings. Reported: log-hazard ratio $\hat{\gamma}_k$, hazard ratio (HR), Wald z -statistic, p -value, and concordance (C-index). Significance threshold: $p < 0.05$.

3 Results

3.1 EBMF

EBMF converged on **20 factors**. Of these, **5 are univariately associated with overall survival** at $p < 0.05$ (Table 1). The strongest association is EBMF6 (C-index = 0.63, $p = 3 \times 10^{-6}$), followed by EBMF4 and EBMF16. EBMF1 has an enormous point estimate ($\text{HR} \approx 4 \times 10^8$) with $p = 0.35$ — a signature of a near-collinear batch factor (very large loadings concentrated in one cohort, unstable Cox fit).

Table 1: Univariate Cox results for each EBMF factor ($K = 20$). Factors with $p < 0.05$ are marked in the final column.

Factor	log-HR	HR	z	p-value	C-index	p < 0.05
EBMF1	33.637	4.058e+14	0.935	0.35	0.536	
EBMF2	0.133	1.142	0.093	0.926	0.526	
EBMF3	3.228	25.22	2.160	0.0308	0.572	
EBMF4	-4.446	0.0117	-2.780	0.00544	0.578	
EBMF5	-1.617	0.1984	-1.082	0.279	0.550	
EBMF6	6.333	562.8	4.653	3e-06	0.628	
EBMF7	0.643	1.902	0.463	0.643	0.507	
EBMF8	-0.805	0.447	-0.585	0.559	0.487	
EBMF9	1.353	3.871	0.928	0.353	0.538	
EBMF10	-0.258	0.7728	-0.188	0.85	0.517	
EBMF11	0.658	1.931	0.421	0.674	0.517	
EBMF12	-0.582	0.5591	-0.385	0.7	0.515	
EBMF13	-2.675	0.0689	-1.730	0.0837	0.544	
EBMF14	-1.414	0.2433	-1.253	0.21	0.521	
EBMF15	0.917	2.503	0.660	0.509	0.515	
EBMF16	3.994	54.24	2.764	0.00571	0.570	
EBMF17	-3.246	0.0389	-2.240	0.0251	0.551	
EBMF18	-0.920	0.3987	-0.643	0.52	0.521	
EBMF19	0.931	2.536	0.674	0.5	0.493	
EBMF20	-0.060	0.9417	-0.042	0.966	0.516	

The cohort-stratified loading heatmap (Figure 1) shows that EBMF separates samples along both a strong cohort-collinear axis (consistent with a platform batch factor) and several axes that mix TCGA and CPTAC samples — the latter being the factors most likely to carry biological rather than technical signal.

Figure 2 annotates the same heatmap with Cox significance: the column strip shows $-\log_{10}(p)$ for each factor, with the active/significant threshold at $-\log_{10}(0.05) \approx 1.3$.

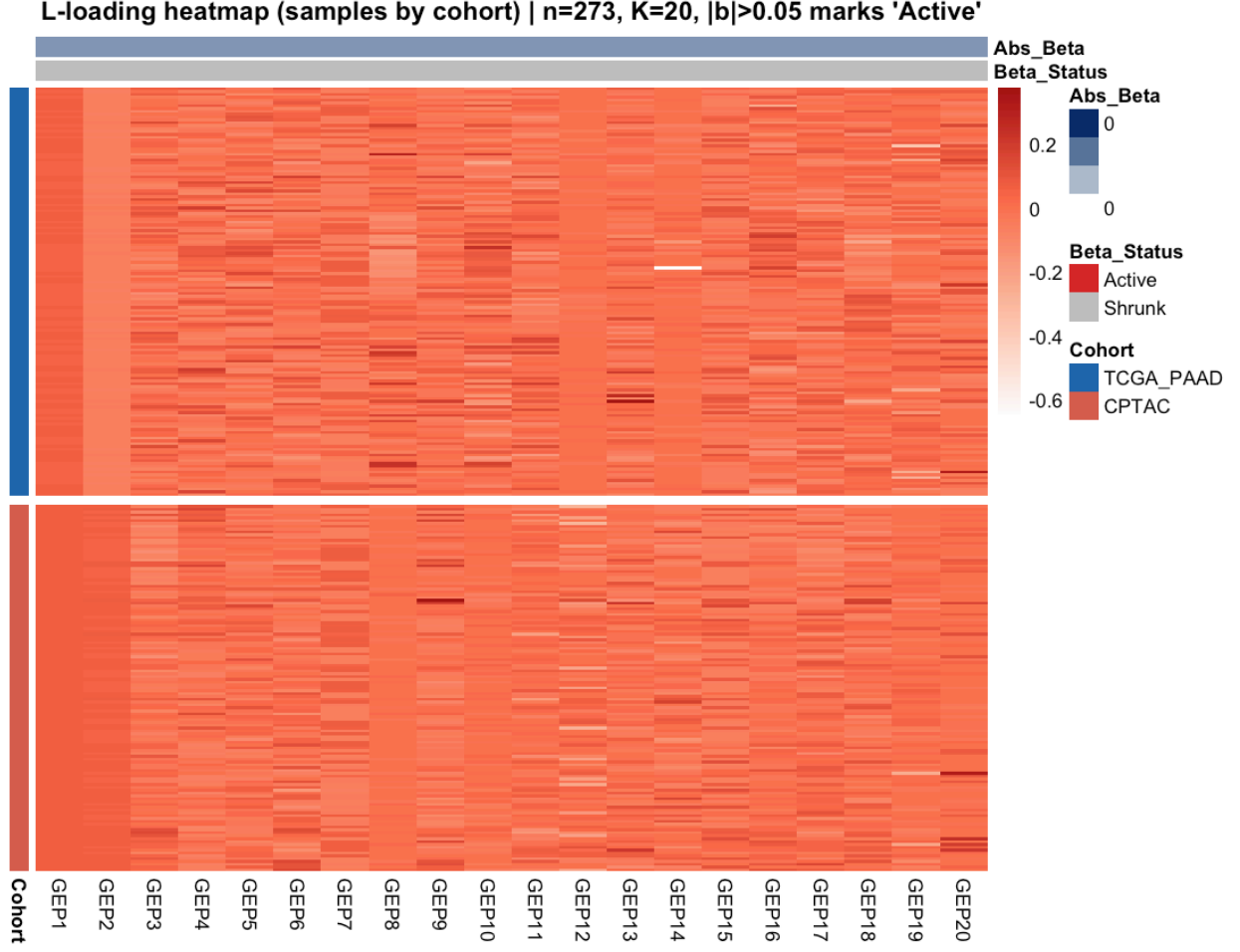


Figure 1: EBMF loading heatmap. Rows = subjects (sorted by cohort), columns = factors GEP1–GEP20. Row annotation (left) = cohort membership. No column annotation is shown because EBMF is unsupervised (is not estimated). Factors with the strongest within-cohort structure are likely batch-driven; factors that mix cohorts are candidates for biologically meaningful signal.

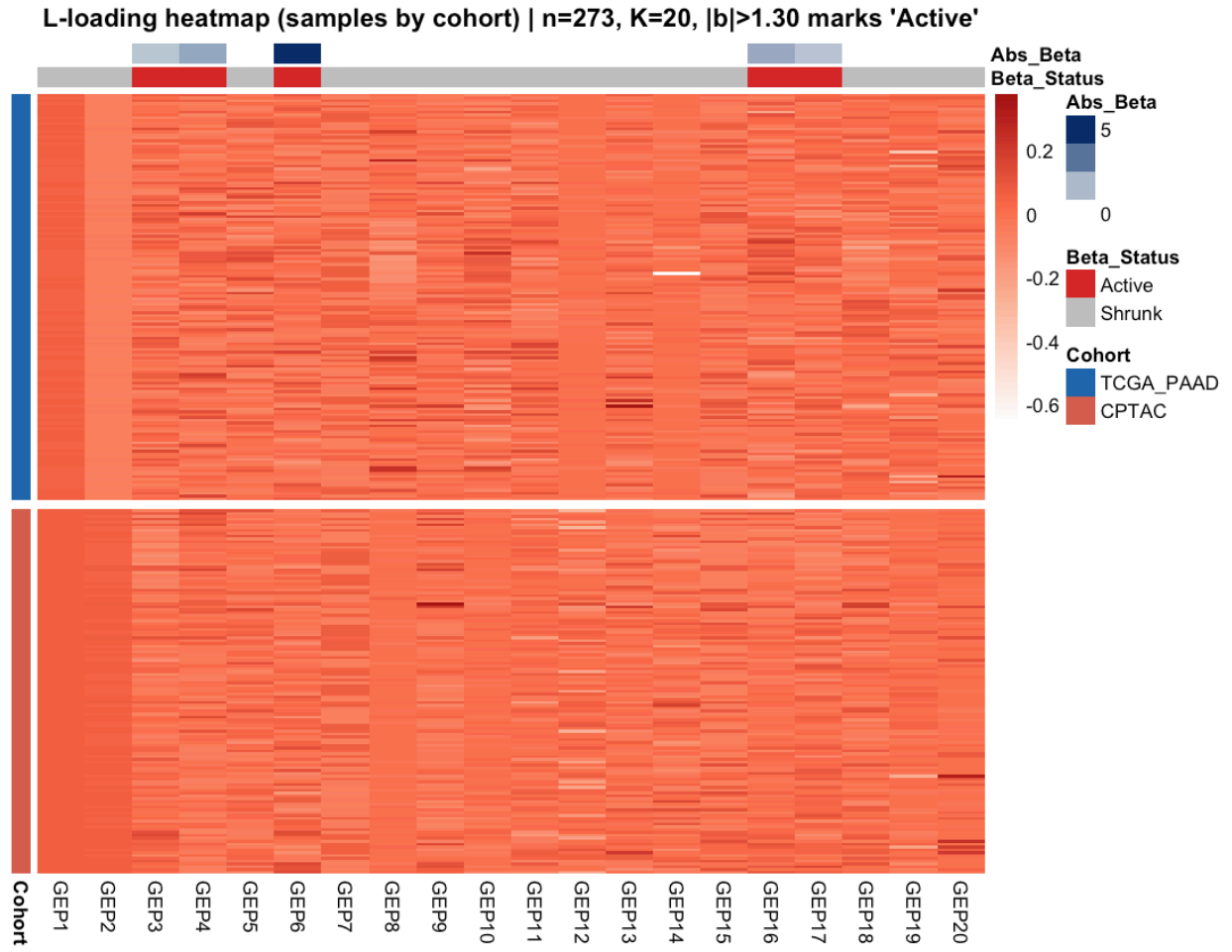


Figure 2: EBMF loading heatmap with Cox-significance annotation. Column annotation strip: orange–blue gradient = $-\log_{10}(\text{p-value})$ from univariate Cox model; ‘Active’ (red) = $p < 0.05$, ‘Shrunk’ (gray) = $p \geq 0.05$.

3.2 PCA

PCA finds **4 of 20 components** significantly associated with survival (Table 2). PC5 is the strongest (C-index = 0.61, $p = 1.5 \times 10^{-4}$). The concordance between EBMF and PCA (5/20 vs. 4/20, overlapping in significance rank) reinforces that the signal is real rather than an artifact of the EBMF model structure.

Table 2: Univariate Cox results for the top 20 PCA components.

Factor	log-HR	HR	z	p-value	C-index	p < 0.05
PC1	0	1	0.107	0.915	0.529	
PC2	0	1	1.905	0.0567	0.565	
PC3	0	0.9999	-2.768	0.00563	0.585	
PC4	0	1	-1.438	0.15	0.538	
PC5	0	1	3.797	0.000147	0.615	
PC6	0	1	-0.150	0.88	0.537	
PC7	0	1	-0.564	0.573	0.537	
PC8	0	1	1.972	0.0486	0.564	
PC9	0	1	-0.663	0.507	0.519	
PC10	0	1	0.959	0.337	0.524	
PC11	0	1	0.156	0.876	0.501	
PC12	0	1	1.882	0.0599	0.543	
PC13	0	1	0.698	0.485	0.534	
PC14	0	0.9999	-1.777	0.0756	0.515	
PC15	0	0.9999	-1.397	0.162	0.541	
PC16	0	1	0.331	0.741	0.517	
PC17	0	1	0.939	0.348	0.525	
PC18	0	1	0.375	0.707	0.494	
PC19	0	0.9998	-2.274	0.023	0.552	
PC20	0	0.9999	-1.007	0.314	0.512	

4 Gene Loadings for Survival-Associated EBMF Factors

The five survival-associated EBMF factors (EBMF3, 4, 6, 16, 17) were inspected for their top gene loadings to characterise the underlying biology. Table 3 lists the top 10 genes by absolute loading for each significant factor.

Table 3: Top 10 genes by |loading| for each survival-associated EBMF factor (K = 5 significant factors shown).

Factor	Rank	Gene	Loading
EBMF16	1	DRD4	-0.1055
EBMF16	2	IAPP	0.0938
EBMF16	3	NPB	-0.0839
EBMF16	4	PSAPL1	0.0800
EBMF16	5	PCSK1N	-0.0784
EBMF16	6	C1QTNF4	-0.0780
EBMF16	7	C1orf194	0.0756
EBMF16	8	BBOX1	0.0756

EBMF16	9	PLAC1	0.0756
EBMF16	10	EMX2OS	-0.0751
EBMF17	1	GPHA2	-0.1600
EBMF17	2	AQP12B	-0.1583
EBMF17	3	APCS	0.1548
EBMF17	4	GUCA1C	-0.1510
EBMF17	5	AQP12A	-0.1486
EBMF17	6	PTF1A	-0.1390
EBMF17	7	FBXW12	-0.1282
EBMF17	8	FGG	0.1270
EBMF17	9	AQP8	-0.1233
EBMF17	10	UMOD	-0.1233
EBMF3	1	GUCA1C	-0.1022
EBMF3	2	PTF1A	-0.0995
EBMF3	3	FGL1	-0.0921
EBMF3	4	GPHA2	-0.0920
EBMF3	5	CELP	-0.0910
EBMF3	6	GSTA2	-0.0855
EBMF3	7	F11	-0.0845
EBMF3	8	UMOD	-0.0834
EBMF3	9	RBPJL	-0.0820
EBMF3	10	AQP12A	-0.0814
EBMF4	1	FCRL1	0.1667
EBMF4	2	TCL1A	0.1662
EBMF4	3	CR2	0.1554
EBMF4	4	FCER2	0.1542
EBMF4	5	CNR2	0.1509
EBMF4	6	PAX5	0.1506
EBMF4	7	FAM129C	0.1497
EBMF4	8	CD19	0.1402
EBMF4	9	MS4A1	0.1375
EBMF4	10	BLK	0.1360
EBMF6	1	A2ML1	0.1164
EBMF6	2	CYP24A1	0.1111
EBMF6	3	BTNL8	-0.1054
EBMF6	4	PGC	-0.1003
EBMF6	5	IVL	0.0999
EBMF6	6	CCL15	-0.0949
EBMF6	7	MUC17	-0.0922
EBMF6	8	PHGR1	-0.0920
EBMF6	9	PSAPL1	-0.0894
EBMF6	10	APOBEC1	-0.0880

Several biologically coherent signals emerge:

- **EBMF3** ($HR > 1$, $p = 0.031$): Strongly negative loadings on *PTF1A*, *GUCA1C*, *FGL1*, *GPHA2*, and *CELP* — exocrine pancreas markers. Higher scores (less exocrine character) associate with worse survival, consistent with the known inverse relationship between exocrine differentiation and PDAC aggressiveness.
- **EBMF4** ($HR < 1$, $p = 0.005$): Top genes *FCRL1*, *TCL1A*, *CR2*, *FCER2*, *CNR2* are canonical B cell / immune markers. Higher factor scores associate with improved survival — a tumour-infiltrating immune signal.

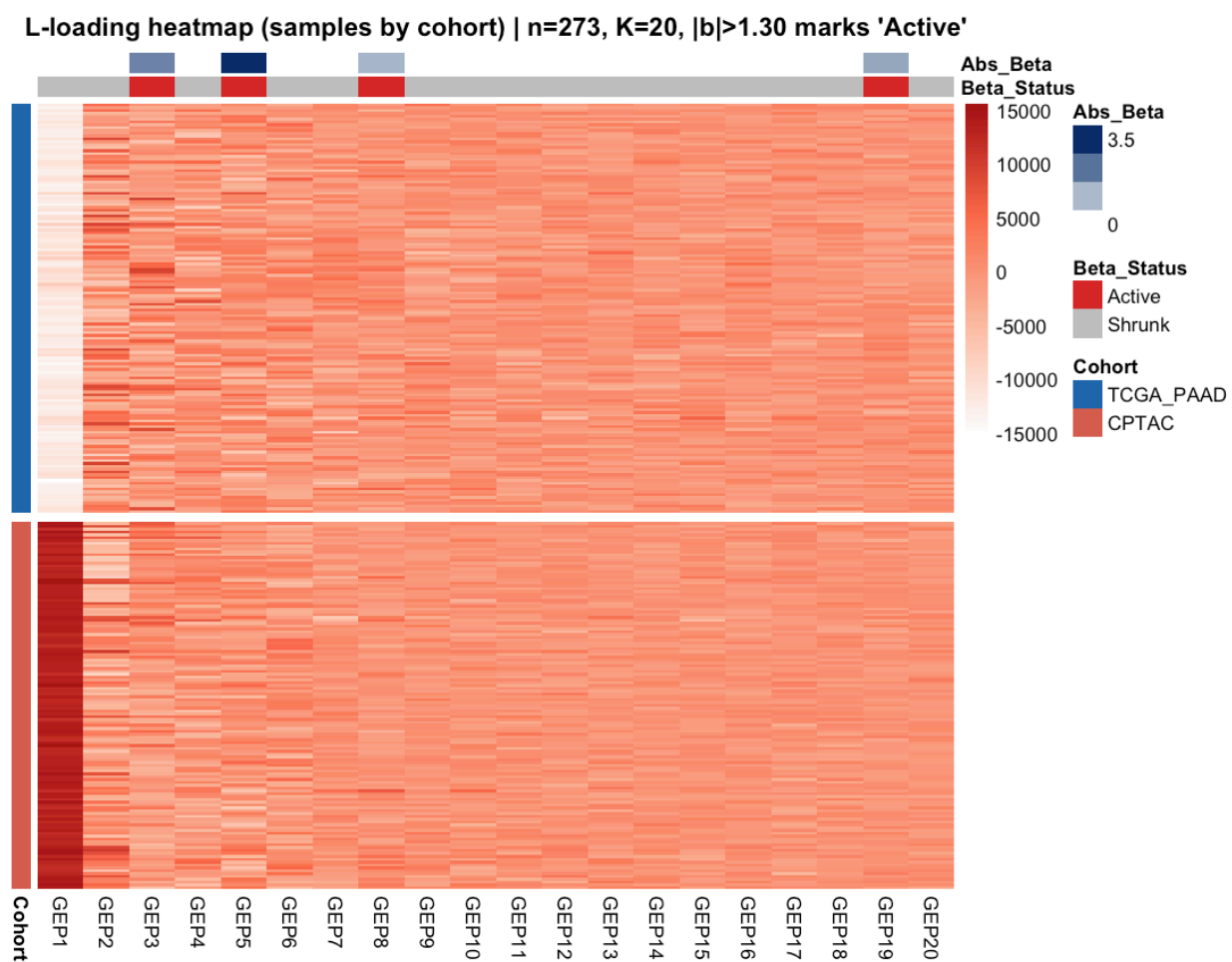


Figure 3: PCA loading heatmap with Cox-significance annotation. Same layout as Figure 2. Column strip = $-\log_{10}(\text{p-value})$; 'Active' = $p < 0.05$.

- **EBMF6** (C-index = 0.63, $p = 3 \times 10^{-6}$, strongest factor): Top genes include *A2ML1*, *CYP24A1*, *CSTA* — a mix of protease-inhibitor, vitamin-D metabolism, and epithelial markers. The mechanism is not immediately interpretable; formal gene-set enrichment analysis is warranted.
- **EBMF16 / EBMF17**: Moderate associations ($p \approx 0.025$); top genes are more mixed, suggesting these may capture subtype heterogeneity rather than a single pathway.

5 Takeaways

The central question is answered: **survival signal exists in the merged v2 data**. Five EBMF factors and four PCA components are univariately associated with overall survival at $p < 0.05$, with the strongest reaching C-index ≈ 0.63 using a single factor’s loadings alone. The failure of SSBMF to assign non-zero $\hat{\beta}$ to any factor is therefore a **model problem**, not a data problem.

Two follow-on observations sharpen the diagnosis:

1. **Lambda increase collapses the model.** A concurrent lambda sweep ($\lambda \in \{1, 5, 10, 20\}$) found that $\lambda = 5$ (and likely higher values) causes the CAVI to degenerate: $\hat{L} \rightarrow 0$ as well as $\hat{\beta} \rightarrow 0$. Naively amplifying the survival objective overshoots and destabilizes the factor-loading updates rather than pulling $\hat{\beta}$ away from zero.
2. **A strong batch factor coexists with survival-associated factors.** EBMF1 has loadings concentrated in one cohort and an unstable Cox estimate — the expected signature of an RNA-seq vs. proteomics platform factor. The fact that *other* factors (EBMF3, 4, 6, 16, 17) simultaneously associate with survival suggests the batch and biological signals occupy distinct directions in the gene space. The challenge for SSBMF is to correctly weight the biological directions under the survival objective without being swamped by the larger-variance batch direction.

Recommended next steps:

- Investigate the CAVI update for **L** to understand why the survival gradient pushes $\hat{L}$ to zero rather than refining it toward the biologically informative factors identified by EBMF.
- Consider **EBMF-initialized SSBMF**: warm-start $\hat{L}$ and $\hat{F}$ from the EBMF solution and only optimize $\hat{\beta}$ from that fixed point. This sidesteps the initialization problem and directly tests whether the β update is capable of selecting the EBMF-identified survival factors.
- Compare factor loadings between EBMF6/EBMF4 (strongest survival association) and the SSBMF GEPs from single-cohort training to understand whether these are the same biological programs or orthogonal signals.